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Dual-polarization rotationally-insensitive monostatic transceiver with standard fiber

Abstract

An apparatus includes polarization beamsplitters that each separate incoming and outgoing optical signals having different polarizations. The apparatus also includes directionally-dependent polarization rotation optical assemblies that each maintain a polarization of one of the incoming and outgoing optical signals and to rotate a polarization of another of the incoming and outgoing optical signals. The apparatus further includes a third polarization beamsplitter that combines the outgoing optical signals to produce transmit optical signals and separate receive optical signals to produce the incoming optical signals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY CLAIM (1) This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Nos. 63/246,648; 63/246,676; and 63/246,693 all filed on Sep. 21, 2021. These provisional applications are hereby incorporated by reference in their entirety. (2) This application is related to the following non-provisional patent applications being filed concurrently herewith: (3) U.S. patent application Ser. No. 17/933,766 filed Sep. 20, 2022 and entitled “SYSTEM AND METHOD FOR DIRECTIONALLY-DEPENDENT POLARIZATION MODIFICATION”; and (4) U.S. patent application Ser. No. 17/933,779 filed Sep. 20, 2022 and entitled “DUAL-POLARIZATION ROTATIONALLY-INSENSITIVE MONOSTATIC TRANSCEIVER WITH DUAL CLADDING FIBER,” now U.S. Pat. No. 12,271,069. (5) These non-provisional applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) This disclosure is generally directed to optical systems. More specifically, this disclosure is directed to a dual-polarization rotationally-insensitive monostatic transceiver with standard fiber.

BACKGROUND

(2) Optical communication systems use light to transport data between locations. Next-generation optical transceivers are being designed for use in free-space optical systems, which refer to systems where optical signals are transmitted and received through free space rather than through optical fibers. Increasing the data capacity of free-space optical systems is becoming more and more important as the demand for high-throughput data products increases. Using two polarizations of light in an optical signal can double the capacity of a single optical channel. However, this is typically achieved using separate transmit and receive apertures in devices using the optical channel, which increases the size, cost, and complexity of the devices. While some monostatic systems have been developed that permit simultaneous transmission and reception of optical signals using a single aperture, these systems rely on the use of multiple wavelengths without any dual polarization capabilities.

SUMMARY

(3) This disclosure is directed to a dual-polarization rotationally-insensitive monostatic transceiver with standard fiber.

(4) In a first embodiment, an apparatus includes first and second polarization beamsplitters each configured to separate incoming and outgoing optical signals having different polarizations. The apparatus also includes first and second directionally-dependent polarization rotation optical assemblies each configured to maintain a polarization of one of the incoming and outgoing optical signals and to rotate a polarization of another of the incoming and outgoing optical signals. The apparatus further includes a third polarization beamsplitter configured to (i) combine the outgoing optical signals to produce transmit optical signals and (ii) separate receive optical signals to produce the incoming optical signals.

(5) In a second embodiment, a system includes a fiber coupling configured to couple to a fiber optic waveguide, where the fiber coupling includes first fiber terminals for a first channel and second fiber couplings for a second channel. The system also includes at least one first polarization beamsplitter configured to separate incoming and outgoing optical signals based on different polarizations of the incoming and outgoing optical signals. The system further includes at least one directionally-dependent polarization rotation optical assembly configured to maintain a polarization of one of the incoming and outgoing optical signals and to rotate a polarization of another of the incoming and outgoing optical signals. The system also includes a second polarization beamsplitter configured to (i) combine the outgoing optical signals from the channels to produce transmit optical

signals and (ii) separate receive optical signals to produce the incoming optical signals for the channels.

(6) In a third embodiment, a method includes separating incoming and outgoing optical signals having different polarizations using first and second polarization beamsplitters. The method also includes passing the incoming optical signals and the outgoing optical signals through first and second directionally-dependent polarization rotation optical assemblies. Each of the directionally-dependent polarization rotation optical assemblies maintains a polarization of one of the incoming and outgoing optical signals and rotating a polarization of another of the incoming and outgoing optical signals. The method further includes combining the outgoing optical signals to produce transmit optical signals and separating receive optical signals to produce the incoming optical signals.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates an example free-space optical (FSO) system according to this disclosure;

(3) FIG. 2 illustrates an example monostatic optical transceiver according to this disclosure; and

(4) FIG. 3 illustrates an example method for transmitting and receiving multiple non-interfering polarization signals according to this disclosure.

DETAILED DESCRIPTION

(5) FIGS. 1 through 3, described below, and the various embodiments used to describe the principles of the present disclosure are by way of illustration only and should not be construed in any way to limit the scope of this disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any type of suitably arranged device or system.

(6) As described above, optical communication systems use light to transport data between locations. Next-generation transceivers are being designed for use in free-space optical systems, which refer to systems where optical signals are transmitted and received through free space rather than through optical fibers. Increasing the data capacity of free-space optical systems is becoming more and more important as the demand for high-throughput data products increases. Using two polarizations of light in an optical signal can double the capacity of a single optical channel. However, this is typically achieved using separate transmit and receive apertures in devices using the optical channel, which increases the size, cost, and complexity of the devices. While some monostatic systems have been developed that permit simultaneous transmission and reception of optical signals using a single aperture, these systems rely on the use of multiple wavelengths without any dual polarization capabilities.

(7) This disclosure provides various implementations of dual-polarization rotationally-insensitive monostatic transceivers. Each dual-polarization rotationally-insensitive monostatic transceiver is configured to simultaneously transmit and receive multiple polarizations, such as on the same wavelength and at the same time, on a non-interfering basis. Moreover, the performance of an optical link between transceivers can remain unaffected by rotational changes between the transceivers at either end of the optical link, which can be achieved passively (without feedback or optical control). Among other things, this is achieved using one or more techniques for using circularly polarized light. As a particular example, using directionally dependent polarization modification of the circularly polarized light can permit an optical signal propagating in one

direction to undergo no polarization change while an optical signal propagating in the opposite direction is rotated by 90°.

(8) Some embodiments of this disclosure provide dual-polarization transmit and receive capabilities in a monostatic transceiver. Various embodiments allow the same wavelength or different wavelengths to be used on any transmit or receiver channel without interference. Single-mode, multi-mode, or PM fiber may be used for outgoing optical signals. When PM fiber is not used, the transceiver may generate orthogonal transmit beams (also referred to as signals) that will not interfere. In some embodiments, a transceiver does not require complex dual cladding fiber and includes a passive rotationally-invariant transmit and receiver capability to simplify use across platforms. Moreover, some embodiments can maintain compatibility with intensity modulator or complex modulation formats. The transceivers can also maintain compatibility with multiple data rates on different channels if needed or desired or maintain compatibility with wavelength division multiplexing (WDM) for increasing data capacity. Some embodiments provide a transceiver that is compatible with an all-fiber-based etalon-enhanced receiver for modularity. Some embodiments enable separate tracking of multiple polarization channels when in the same field of view. According to some embodiments, position, acquisition, and tracking flexibility can be used depending on how a transceiver is configured. In addition, a monostatic transceiver may use directionally-dependent polarization modification to apply relative polarization rotation as a function of a direction of propagation of a light beam.

(9) FIG. 1 illustrates an example free-space optical (FSO) system **100** according to this disclosure. As shown in FIG. 1, the system **100** includes two nodes **102** and **104** that communicate with one another optically. Each node **102** and **104** represents a ground-, air-, or space-based system that can transmit or receive data using optical communications. In this example, the nodes **102** and **104** can engage in bidirectional communication with one another. However, this is not necessarily required, and the nodes **102** and **104** may engage in unidirectional communication with one another (meaning one node **102** or **104** may only transmit and the other node **104** or **102** may only receive, at least with respect to each other).

(10) The node **102** in this example includes an optical transmitter **106**, which generally operates to produce optical signals **108** used for communication or other purposes. For example, the optical transmitter **106** may encode information onto the optical signals **108**, such as by using suitable amplitude, phase, frequency, or other modulation(s) of light. The optical signals **108** can be transmitted through free space or other transmission medium to the node **104**, where an optical receiver **110** receives and processes the optical signals **108**. For instance, the optical receiver **110** can identify the amplitude, phase, frequency, or other modulation(s) of light in the optical signals **108** and use the identified modulation(s) to recover the information encoded onto the optical signals **108**. Any suitable type of modulation/demodulation scheme may be used here to encode and decode the optical signals **108** (assuming communication is one purpose of the optical signals **108**). Since the nodes **102** and **104** are bidirectional in this example, the same process can be used in the opposite direction, meaning an optical transmitter **112** of the node **104** produces optical signals **114** that are transmitted towards the node **102** and received and processed by an optical receiver **116** of the node **102**.

(11) Note that while the optical transmitter **106** and the optical receiver **116** are shown here as separate components, they can be integrated into a single optical transceiver **118**. This may allow, for example, the same structure to be used for both transmission and reception purposes. Similarly, while the optical transmitter **112** and the optical receiver **110** are shown here as separate components, they can be integrated into a single optical transceiver **120**. This may allow, for instance, the same structure to be used for both transmission and reception purposes.

(12) The optical transmitters, receivers, and transceivers described in this disclosure may find use in a large number of applications, such as communications, imaging, and remote sensing applications. For example, optical transmitters, receivers, or transceivers may be used in data

centers or telecommunication systems to transport information rapidly between locations, including the transport of large amounts of information over very large distances. Optical transmitters, receivers, or transceivers may be used in consumer or commercial electronic devices, biomedical devices, or advanced computing devices to support optical-based communications with those devices. Optical transmitters, receivers, or transceivers may be used in airplanes, drones, satellites, autonomous vehicles, rockets, missiles, or other commercial or defense-related systems. Optical transmitters, receivers, or transceivers may be used in non-communication-related optical applications, such as laser detection and ranging (LADAR) applications or other applications that can use polarimetric processing. In general, this disclosure is not limited to any particular application of the optical transmitters, receivers, and transceivers.

(13) Although FIG. 1 illustrates one example of a free-space optical system **100**, various changes may be made to FIG. 1. For example, while only two nodes **102** and **104** are shown here, the system **100** may include any suitable number of nodes that engage in any suitable unidirectional, bidirectional, or other communications or other interactions with each other. Also, each node of the system **100** may include any suitable number of optical transmitters, receivers, or transceivers that communicate or otherwise use any number of optical signals. In addition, the system **100** is shown in simplified form here and may include any number of additional components in any suitable configuration as needed or desired.

(14) FIG. 2 illustrates an example monostatic optical transceiver **200** according to this disclosure. The optical transceiver **200** may, for example, represent or be used within the optical nodes **102** and **104** of the system **100** of FIG. 1. However, the optical transceiver **200** may be used in any other suitable device(s) and in any other suitable system(s).

(15) As shown in FIG. 2, the monostatic optical transceiver **200** includes a collection telescope **202**, a quarter waveplate **204**, a polarization beam splitter (PBS) **206**, and at least one directionally-dependent optical assembly (DDOA) **208a-208b**. Each NROA **208a-208b** includes a Faraday rotator **210** and a waveplate **212**. Each DDOA **208a-208b** represents a directionally-dependent polarization rotation optical assembly. The waveplate **212** can be a half waveplate or a quarter waveplate. In some embodiments, each DDOA **208a-208b** includes a Faraday rotator **210** providing a polarization rotation of about 45° and a half waveplate **212** with its fast axis oriented at about 22.5° from the +x axis. However, other embodiments of the Faraday rotator **210** and the waveplate **212** may be used. For instance, various arrangements of Faraday rotators and waveplates are described in U.S. patent application Ser. No. 17/933,766 filed on Sep. 20, 2022 and entitled "SYSTEM AND METHOD FOR DIRECTIONALLY-DEPENDENT POLARIZATION MODIFICATION," which has been incorporated by reference above. In some embodiments, the monostatic optical transceiver **200** includes two or more channels and is configured to transmit and receive on each channel. Here, the monostatic optical transceiver **200** can employ two or more DDOA **208a-208b** to provide directionally-dependent polarization modifications, such as to permit a beam propagating in one direction to undergo no substantial polarization change while a beam propagating in the opposite direction is rotated by about 90° .

(16) In the optical transceiver **200**, optical communication signals can be transmitted and received via respective light beams, which carry communication signals. Note that the terms "beams" and "signals" may be used interchangeably. In some cases, optical communication signals can be transmitted and received on at least two wavelengths or channels, where each channel includes transmit optical signals and receive optical signals. The collection telescope **202** is configured to receive and collimate incoming receive (Rx) beams **214** and diverge and direct outgoing transmit (Tx) beams **216**. Each channel may include at least one transmit beam **216** and at least one receive beam **214**.

(17) The quarter waveplate **204** is configured, for each channel, to convert a polarization of light from a linear polarization to a circular/elliptical polarization and vice versa. For example, on a first channel (Ch1), the quarter waveplate **204** may be configured to (i) convert a transmit beam **218**

having a P linear polarization (transmit PLP) into a transmit beam **216** having a right-hand circular (RHC) polarization and (ii) convert a receive beam **214** having a right-hand circular (RHC) polarization into a receive beam **220** having a P linear polarization (receive PLP). On a second channel (Ch2), the quarter waveplate **204** may be configured to (i) convert another transmit beam **218** having an S linear polarization (transmit SLP) into another transmit beam **216** having a left hand (LHC) polarization and (ii) convert another receive beam **214** having an LHC polarization into another receive beam **220** having an S linear polarization (receive SLP). The ability to transmit and receive optical beams having circular polarizations enables devices to communicate with one another irrespective of orientation. As a result, the quarter waveplate **204** enables two nodes (such as nodes **102** and **104** in FIG. 1) to communicate without having to be aligned with each other. In this example, the transmit beam **216** can include signals with P and S linear polarizations, and the receive beam **214** can include signals with P and S linear polarizations.

(18) The PBS **206** is configured to combine and separate channels by combining and separating P and S components within optical beams. For example, in the outgoing direction, transmit beams are received from standard fiber terminals **232** and **234**. The fiber terminals **232** and **234** are associated with different channels and can represent at least part of a fiber coupling that is configured to be coupled to at least one fiber optic cable. The PBS **206** can combine transmit beams **218a-218b** having different P and S polarizations in different channels into the transmit beams **218**. In the incoming direction, the PBS **206** can separate the receive beams **220** into receive beams **220a-220b** having different P and S polarizations in the different channels. In this example, the transmit beam **218a** and the receive beam **220a** both have a P linear polarization, and the transmit beam **218b** and the receive beam **220b** both have an S linear polarization. As can be seen here, the PBS **206** is configured to separate the channels for the receive beams **220** and combine the channels for the transmit beams **218**. In this example, a mirror or other reflector **222** can be used to help direct signals to and from the PBS **206**.

(19) Each NROA **208a-208b** can be configured to maintain or modify light according to a direction of propagation of the light through the NROA **208a-208b**. In the example shown in FIG. 2, the NROA **208a** receives the transmit beam **218a**, and the NROA **208b** receives the transmit beam **218b**. In some cases, each NROA **208a-208b** is configured to maintain an angular polarization of the associated transmit beam **218a-218b** while modifying an angular polarization of an associated receive beam **220a-220b**. For example, the transmit beams **218a-218b** propagate through the respective NROAs **208a-208b** in a first direction, while the receive beams **220a-220b** propagate through the NROAs **208a-208b** in a second direction opposite the first direction. In some cases, the transmit beams **218a-218b** may pass through the NROAs **208a-208b** with little or no polarization modifications, while the receive beams **220a-220b** can have their polarizations rotated as they propagate through the NROAs **208a-208b**. This results in the creation of receive beams **220a'** and **220b'**. Accordingly, each incoming beam can propagate along the receive direction and have its polarization rotated by about 90° or other angle compared to the associated outgoing beam (which may remain unchanged). Note, however, that this can be reversed so that the outgoing signals are rotated and the incoming signals are not. This approach supports the use of dual polarizations in the optical transceiver **200**, which helps to provide rotational invariance between terminals with quarter waveplates. In some embodiments, the optical signals in the different channels of the optical transceiver **200** have different wavelengths.

(20) In this example, a PBS **224** is configured to pass the transmit beam **218a** and reflect the receive beam **220a'**, and a PBS **228** is configured to pass the receive beam **220b'** and reflect the transmit beam **218b**. Among other things, this directs the receive beams **220a'-220b'** to standard fiber terminals **226** and **230**. Note, however, that the passing and reflectings may be reversed here. The fiber terminals **226** and **230** are associated with different channels and can represent at least part of a fiber coupling that is configured to be coupled to at least one fiber optic cable.

(21) Note that the depictions of the signals, polarizations, and fibers as shown in FIG. 2 are for

illustration only, and the particulars of the separation and combining of signals can be changed without departing from the scope of this disclosure. Embodiments of the present disclosure provide a monostatic optical transceiver **200** that include at least one non-reciprocal optical assembly, namely one or more NROAs **208a-208b**, which are used to provide directionally-dependent polarization modification that enable multiple beams to be used in the same transceiver aperture. In this example, the optical transceiver **200** is configured to communicate in free space using light having circular polarizations, which makes devices insensitive to relative orientation. The optical transceiver **200** here is designed so that counter-propagating beams of the same polarization can be separated on a non-interfering basis to provide the use of dual polarization.

(22) In some embodiments, as described above, the NROAs **208a-208b** can be configured to pass light in one direction substantially unmodified and to rotate the polarization of light passing in the opposite direction (such as by 90°). In some cases, this can be achieved using a Faraday rotator **210** that provides a polarization rotation of 45° and a half waveplate **212** with its fast axis oriented at 22.5° from the $+x$ axis. In the first channel Ch1, transmit light (P polarized) is rotated 45° by the Faraday rotator **210** and then— 45° by the half waveplate **212**, leaving the outgoing polarization substantially unchanged. In the second channel Ch2, transmit light (S polarized) is rotated to 135° by the Faraday rotator **210**, leaving an angle of 67.5° between the half waveplate's axis and the incident light. After passing through the half waveplate, the transmit light is rotated back, again leaving the outgoing polarization substantially unchanged. The outgoing beams are combined by the PBS **206**, and the S and P polarizations are converted to circular polarizations after passing through the quarter waveplate **204**. This enables dual-channel transmission.

(23) For dual-channel rotationally-invariant reception, circularly-polarized light is captured by the collection telescope **202** and provided to the quarter waveplate **204**, which can have its fast axis rotated by -45° (as viewed from the incoming light). The incoming light is converted to S and P polarized light, and these two orthogonal polarizations are separated by the PBS **206** into separate paths. In these paths, the S and P polarized receive beams overlap the S and P polarized transmit beams in space, but the transmit and receive beams travel in opposite directions in each path. In the NROA **208a**, the P polarized receive beam is rotated to $+135^\circ$ by the half waveplate **212** and to the S polarization by the Faraday rotator **210**. In the NROA **208b**, the S polarized receive beam is rotated $+45^\circ$ by the half waveplate **212** and to the P polarization by the Faraday rotator **210**. When propagating along the receive direction, both polarizations are rotated by 90° (while the transmit beams may be unchanged). This allows for dual polarization reception with rotational invariance. Note, however, that the positions of the Faraday rotator **210** and the waveplate **212** can be reversed.

(24) Although FIG. 2 illustrate one example of a monostatic optical transceiver **200**, various changes may be made to FIG. 2. For example, the monostatic optical transceiver **200** may include any suitable number of optical devices or different types of devices. As a more specific example, the particulars of the separation and combination of beams can be changed. Also, while the monostatic optical transceiver **200** shown in FIG. 2 is described as using half waveplates **212** in the NROAs **208a-208b**, embodiments using quarter waveplates can be used. In addition, any other suitable components may be used with the monostatic optical transceiver **200** to support any other desired functions of the monostatic optical transceiver **200**.

(25) FIG. 3 illustrates an example method **300** for transmitting and receiving multiple non-interfering polarization signals according to this disclosure. For ease of explanation, the method **300** is described with respect to the monostatic optical transceiver **200** of FIG. 2. However, the method **300** could be used with any other suitable optical transceiver.

(26) In step **302**, transmit beams and receive beams are communicated via one or more waveguides, such as one or more standard fiber-optic cables. For example, two transmit beams can be received via fiber terminals **232** and **234** from one or more standard fiber-optic cables, and two receive beams can be provided via fiber couplings to one or more standard fiber-optic cables. Each of the transmits beams can be received with a respective polarity and on a respective channel, and

each of the receive beams can be provided with a respective polarity on a respective channel. As a particular example, a first transmit beam with a P polarization may be received on a first channel, a second transmit beam with an S polarization may be received on a second channel, a first receive beam with an S polarization may be provided on the first channel, and a second receive beam with a P polarization may be provided on the second channel.

(27) In step **304**, the transmit and receive beams for each channel are isolated. For example, the transmit and receive beams for the first channel may be separated by the PBS **224**, and the transmit and receive beams for the second channel may be separated by the PBS **228**. Each PBS **224** and **228** can separate light based on the polarizations in the light. In step **306**, the transmit and receive beams are propagated through non-reciprocating optical assemblies, such as NROAs **208a-208b**. For example, in the first channel, the first transmit beam and the first receive beam can propagate through a first NROA **208a**. In the second channel, the second transmit beam and the second receive beam can propagate through a second NROA **308b**. In some cases, each NROA **208a-208b** may be configured to pass light (such as the transmit beams) propagating in one direction substantially unmodified while rotating the polarization of light (such as the receive beams) propagating in the opposite direction.

(28) In step **308**, the transmit beams from the channels are combined, and the receive beams are separated into the channels. For example, the transmit beams from the channels may be combined by allowing the transmit beam from the first channel to propagate through the PBS **206** while reflecting the transmit beam from the second channel by the PBS **206**. The PBS **206** can also split incoming light into the receive beams in the channels based on the polarizations in the incoming light.

(29) In step **310**, the transmit and receive beams propagate through a quarter waveplate. The quarter waveplate **204** converts the transmit beams from linear polarizations to circular/elliptical polarizations, and the quarter waveplate **204** converts the receive beams from circular/elliptical polarizations to linear polarizations. In step **312**, the beams are used for communication or other purposes.

(30) Although FIG. **3** illustrates one example of a method **300** for transmitting and receiving multiple non-interfering polarization signals, various changes may be made to FIG. **3**. For example, while shown as a series of steps, various steps in FIG. **3** could overlap, occur in parallel, occur in a different order, or occur multiple times.

(31) It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

(32) The description in the present disclosure should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims invokes 35 U.S.C. § 112(f) with respect to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the

features of the claims themselves, and is not intended to invoke 35 U.S.C. § 112(f).

(33) While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. An apparatus comprising: first and second polarization beamsplitters each configured to separate optical signals having different polarizations, wherein the optical signals having the different polarizations comprise incoming optical signals that pass through an aperture and outgoing optical signals that pass through the aperture; first and second directionally-dependent polarization rotation optical assemblies each configured to maintain a polarization of one of the optical signals and to rotate a polarization of another of the optical signals; and a third polarization beamsplitter configured to (i) combine the outgoing optical signals to produce transmit optical signals and (ii) separate receive optical signals to produce the incoming optical signals.
2. The apparatus of claim 1, wherein the first directionally-dependent polarization rotation optical assembly comprises: a Faraday rotator and a waveplate optically positioned between the first polarization beamsplitter and the third polarization beamsplitter, the Faraday rotator and the waveplate collectively configured to provide a relative polarization change between (i) light propagating in a first direction through the Faraday rotator and the waveplate and (ii) light propagating in a second direction opposite the first direction through the Faraday rotator and the waveplate.
3. The apparatus of claim 1, wherein the third polarization beamsplitter is configured to: combine a first outgoing optical signal of the outgoing optical signals from a first channel with a second outgoing optical signal of the outgoing optical signals from a second channel; and provide a first incoming optical signal of the incoming optical signals for the first channel and a second incoming optical signal of the incoming optical signals for the second channel.
4. The apparatus of claim 3, wherein: the first directionally-dependent polarization rotation optical assembly is configured to rotate a polarization of the first incoming optical signal from a P polarization to an S polarization while maintaining a P polarization of the first outgoing optical signal; and the second directionally-dependent polarization rotation optical assembly is configured to rotate a polarization of the second incoming optical signal from an S polarization to a P polarization while maintaining an S polarization of the second outgoing optical signal.
5. The apparatus of claim 1, further comprising: a fiber coupling configured to couple to a fiber optic cable, the fiber coupling comprising first fiber terminals for a first channel and second fiber terminals for a second channel.
6. The apparatus of claim 5, wherein: the first channel is associated with a first outgoing optical signal of the outgoing optical signals and a first incoming optical signal of the incoming optical signals, the first outgoing optical signal and the first incoming optical signal associated with a first wavelength; and the second channel is associated with a second outgoing optical signal of the outgoing optical signals and a second incoming optical signal of the incoming optical signals, the second outgoing optical signal and the second incoming optical signal associated with a second wavelength.
7. The apparatus of claim 1, further comprising: a waveplate configured to (i) convert the transmit optical signals into first free-space optical signals having a right-hand circular polarization to be transmitted on a first wavelength and a left-hand circular polarization to be transmitted on a second wavelength and (ii) convert second free-space optical signals having a left-hand circular polarization received on the first wavelength and a right-hand circular polarization received on the

second wavelength into the receive optical signals having a linear polarization.

8. The apparatus of claim 7, further comprising a transceiver; wherein the transceiver comprises the polarization beamsplitters, the directionally-dependent polarization rotation optical assemblies, and the waveplate.

9. A system comprising: a fiber coupling configured to couple to a fiber optic waveguide, the fiber coupling comprising first fiber terminals for a first channel and second fiber terminals for a second channel; at least one first polarization beamsplitter configured to separate optical signals having different polarizations, wherein the optical signals having the different polarizations comprise incoming optical signals that pass through the fiber optic waveguide and outgoing optical signals that pass through the fiber optic waveguide; at least one directionally-dependent polarization rotation optical assembly configured to maintain a polarization of one of the optical signals and to rotate a polarization of another of the optical signals; and a second polarization beamsplitter configured to (i) combine the outgoing optical signals to produce transmit optical signals and (ii) separate receive optical signals to produce the incoming optical signals.

10. The system of claim 9, wherein the at least one directionally-dependent polarization rotation optical assembly comprises: a Faraday rotator and a waveplate collectively configured to provide a relative polarization change between (i) light propagating in a first direction through the Faraday rotator and the waveplate and (ii) light propagating in a second direction opposite the first direction through the Faraday rotator and the waveplate.

11. The system of claim 10, wherein the Faraday rotator and the waveplate are collectively configured to apply a 90° polarization rotation to the light propagating in one of the first and second directions.

12. The system of claim 9, wherein the second polarization beamsplitter is configured to: combine a first outgoing optical signal of the outgoing optical signals from the first channel with a second outgoing optical signal of the outgoing optical signals from the second channel; and provide a first incoming optical signal of the incoming optical signals for the first channel and a second incoming optical signal of the incoming optical signals for the second channel.

13. The system of claim 9, further comprising: a waveplate configured to (i) convert one or more linear polarizations of the transmit optical signals into one or more circular/elliptical polarizations and (ii) convert one or more circular/elliptical polarizations of the receive optical signals into one or more linear polarizations.

14. The system of claim 9, further comprising: a telescope configured to emit the transmit optical signals and receive the receive optical signals.

15. A method comprising: separating optical signals having different polarizations using first and second polarization beamsplitters, wherein the optical signals having the different polarizations comprise incoming optical signals that pass through an aperture and outgoing optical signals that pass through the aperture; passing the optical signals through first and second directionally-dependent polarization rotation optical assemblies, each of the directionally-dependent polarization rotation optical assemblies maintaining a polarization of one of the optical signals and rotating a polarization of another of the optical signals; combining the outgoing optical signals to produce transmit optical signals; and separating receive optical signals to produce the incoming optical signals.

16. The method of claim 15, wherein the first directionally-dependent polarization rotation optical assemblies comprises: a Faraday rotator and a waveplate collectively providing a relative polarization change between (i) light propagating in a first direction through the Faraday rotator and the waveplate and (ii) light propagating in a second direction opposite the first direction through the Faraday rotator and the waveplate.

17. The method of claim 16, wherein a third polarization beamsplitter combines the outgoing optical signals to produce the transmit optical signals and separates the receive optical signals to produce the incoming optical signals.

18. The method of claim 15, wherein: a first channel is associated with a first outgoing optical signal of the outgoing optical signals and a first incoming optical signal of the incoming optical signals, the first outgoing optical signal and the first incoming optical signal associated with a first wavelength; and a second channel is associated with a second outgoing optical signal of the outgoing optical signals and a second incoming optical signal of the incoming optical signals, the second outgoing optical signal and the second incoming optical signal associated with a second wavelength.

19. The method of claim 18, wherein: the first directionally-dependent polarization rotation optical assembly rotates a polarization of the first incoming optical signal from a P polarization to an S polarization while maintaining a P polarization of the first outgoing optical signal; and the second directionally-dependent polarization rotation optical assembly rotates a polarization of the second incoming optical signal from an S polarization to a P polarization while maintaining an S polarization of the second outgoing optical signal.

20. The method of claim 15, further comprising: using a waveplate to: convert the transmit optical signals into first free-space optical signals having a right-hand circular polarization to be transmitted on a first wavelength and a left-hand circular polarization to be transmitted on a second wavelength; and convert second free-space optical signals having a left-hand circular polarization received on the first wavelength and a right-hand circular polarization received on the second wavelength into the receive optical signals having a linear polarization.
